MFE230Z

From equity pairs to options relative value

Learned sizing of daily delta-hedged, initially vega-neutral downside puts

1.003 walk-forward excess Sharpe at $10m

- Adam Kreitzman and Soo Xin Yang
- 2018–29 August 2025 · Actual option contracts · Research draft

MOTIVATING PAPER

Ning & Lee: formation and trading are separate problems

Advanced Statistical Arbitrage with Reinforcement Learning (2024)

- Formation: empirical mean reversion time (EMRT) selects spread coefficients.
- Trading: a Q-policy responds to recent spread movements.
- Ten equity pairs: formed in 2022, traded in 2023.

$$X = SA - B\,SB$$

ORIGINAL PAPER

What the Q-learner learns

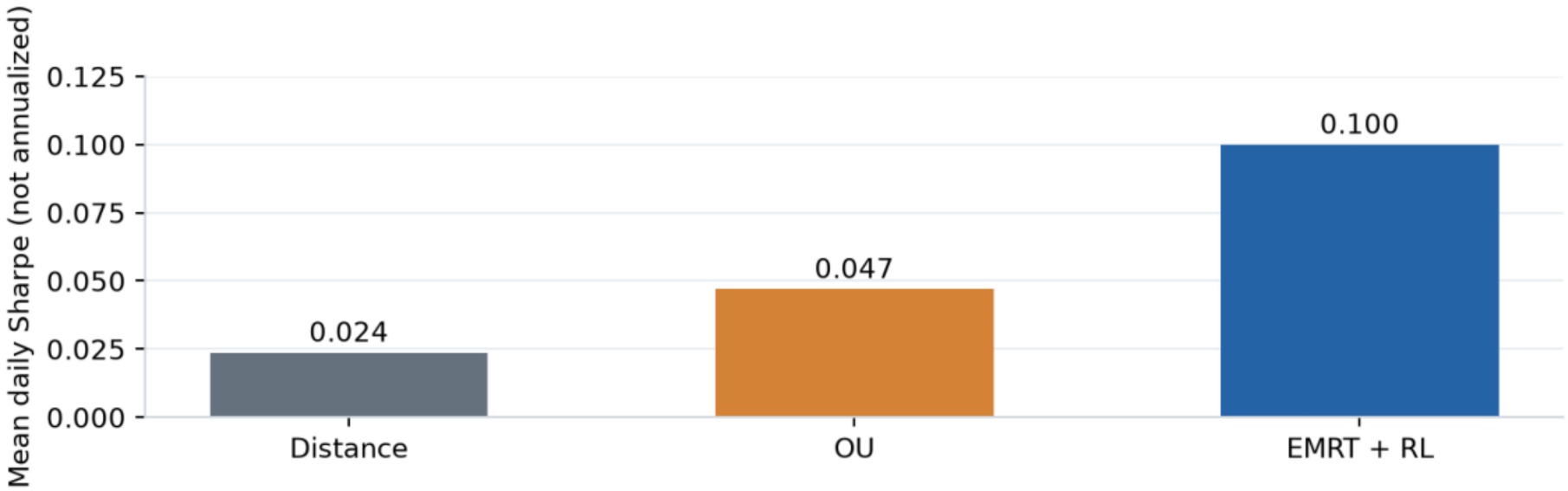
Synthetic mean-reverting training; real equity evaluation

- Recent changes become discrete states; actions buy, hold or sell under position restrictions.
- Reward uses deviation from the true spread mean, minus transaction cost.
- Synthetic paths supply that known mean; transfer to changing market regimes remains a question.

PUBLISHED EVIDENCE

The paper reports higher daily Sharpe in all ten pairs

Keep its daily metric separate from our annualized excess Sharpe



Ning and Lee (2024), Tables 3–4: arithmetic mean of reported daily Sharpe across ten pairs, formed in 2022 and traded in 2023. These published equity results use a different metric and setup from our annualized options results.

CRITICISM → RESEARCH DESIGN

What we need to test in the volatility extension

Longer evaluation, economically correct rewards and controlled sizing comparisons

- One equity trading year does not establish robustness across regimes.
- Formation and trading change together; isolate learned sizing with common proposals.
- IV-spread convergence is not the same as profitable delta-hedged options.
- Use contract marks, hedges, financing, costs and observable inputs.

OUR RESEARCH PATH

ATM proxy signal did not transfer unchanged

We moved from approximate ATM payoff to actual contracts and separate smile ideas

Stage	Finding	Meaning
ATM IV proxy	Sized 1.461; NN 1.036	Silver concentrated; full-size policy 1.497
Actual ATM calls	Sentiment checkpoint 0.368	Daily funded cash-excess results
P25 puts + learned size	1.006 reference; 1.003 at $10m	Retained contract-level candidate

UNIVERSE AND DATA

Broad research universe; sparse eligible trades

345 candidate pair sleeves → 608 proposals → 212 entries / 42 pairs at $10m

- Actual option IDs, quotes, Greeks, underlying histories, dividends and rates.
- Inputs-only P25 CSV: 36,327 rows; 33 feature columns plus identifiers and eligibility.
- Monthly completed-history fits; no forward reward is an action input.
- Inherited retrospective pair calendars; no untouched holdout.

RETAINED TRADE

Approximately 30-day, 25-delta puts

P25 is downside volatility; neutrality is imposed at entry



Retained P25 strategy: rules propose the direction; the learner adjusts size. Orders use prior-close selected contracts on the next session. Positions hold to each leg’s expiry, except audited corporate exits.

- Same option type; integer contract counts match initial vega within 10%.
- Daily stock delta hedge; each leg held to expiry except audited corporate exits.

EXECUTION AND RISK

Account for actual positions and costs

A shared cash-and-stock ledger, independently reconciled

Item	Retained assumption
Option execution	$0.03 / contract + 0.10 vol-point entry penalty per leg
Stock hedging	Daily delta hedge; 1 bp turnover cost
Training labels	$0.65 / contract hurdle
Risk admission	5% stress; 80% reserve; 2× gross stock hedge

- Reserves are internal estimates, not broker margin. Held positions can breach limits.
- No early assignment, forced liquidation or nonlinear market impact.

TRADE SIGNAL

Keep relative value separate from size allocation

Frozen direction rule; learned 50 / 75 / 100% vega budget

- 252-session IV-spread rank: lower / upper quintile.
- Backward beta 0.85–1.15 and return correlation ≥ 0.5.
- Positive completed 91 / 365-day directional memory; count minima 21 / 63.
- Full-size reference: $450 vega per vol point per leg.

ECONOMIC STATE

Relative expected variance premium improves the state

27 existing inputs + three economic quantities and availability flags

$$VRP = IV^2 - \text{estimated variance}$$

$$\text{estimated variance} = 252 \times EWMA20(r^2)$$

- Long-minus-short EWMA premium, prior-history premium z-score and realized/implied variance ratio.
- 20-session half-life; historical normalization uses strictly prior 252 sessions.
- Existing IV / RV, beta, correlation, DTE, VIX and profitability memory remain.

LEARNING MECHANISM

A contextual sizing policy optimized for daily Sharpe

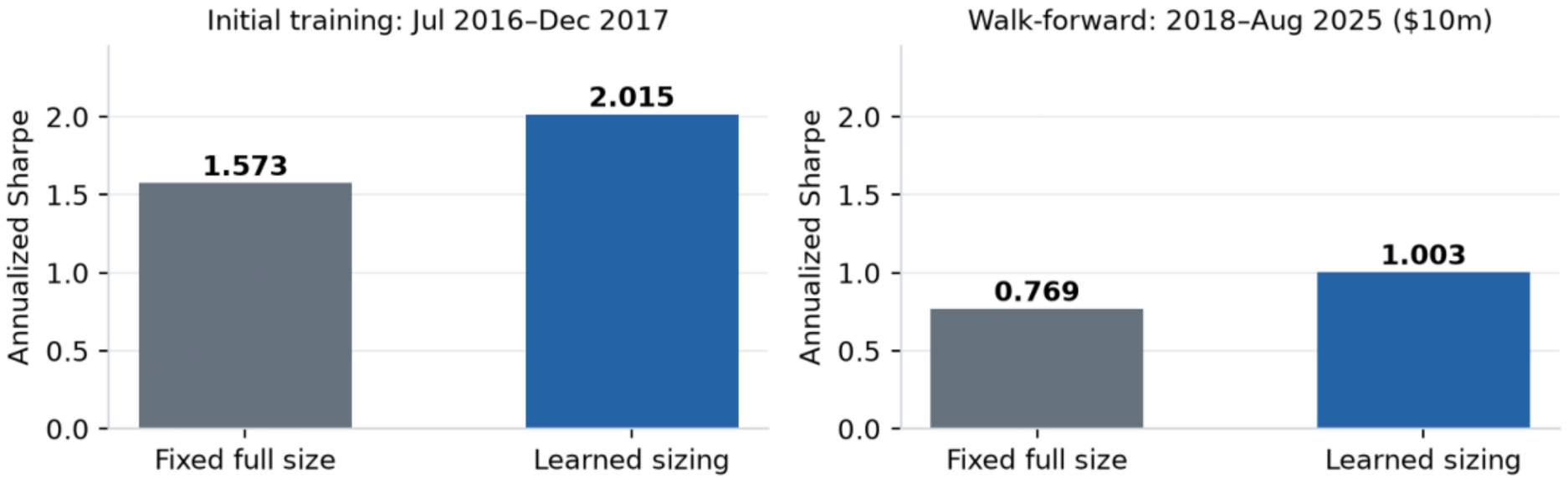
Regression tree → discrete states → CEM size allocation

- At most 8 leaves; actions 50 / 75 / 100%.
- 50 search iterations × 192 candidate allocations; joint training P&L covariance.
- Rolling 18-month completed-outcome training; monthly refits.
- Direct contextual policy search—not Q-learning or a neural actor-critic.

TRAINING AND EVALUATION

Sizing benefit appears before and after 2018

Within-period comparisons; different training and evaluation accounting

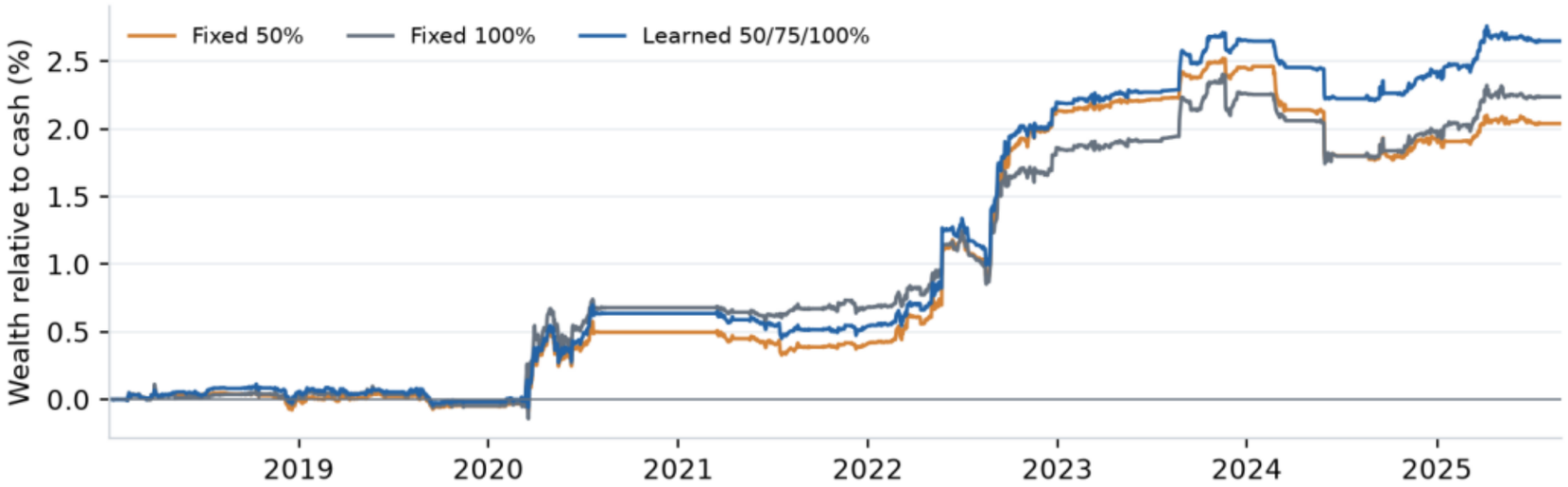


Learned sizing improves each within-period benchmark. Training uses aggregated P&L paths and 0.65*commissions*; *walk − forwardusesthefundedledgerand*0.03 commissions. The levels are not directly comparable.

MAIN PORTFOLIO RESULT

Learned sizing improves the fixed-vega benchmarks

2018–August 2025 at $10m equity



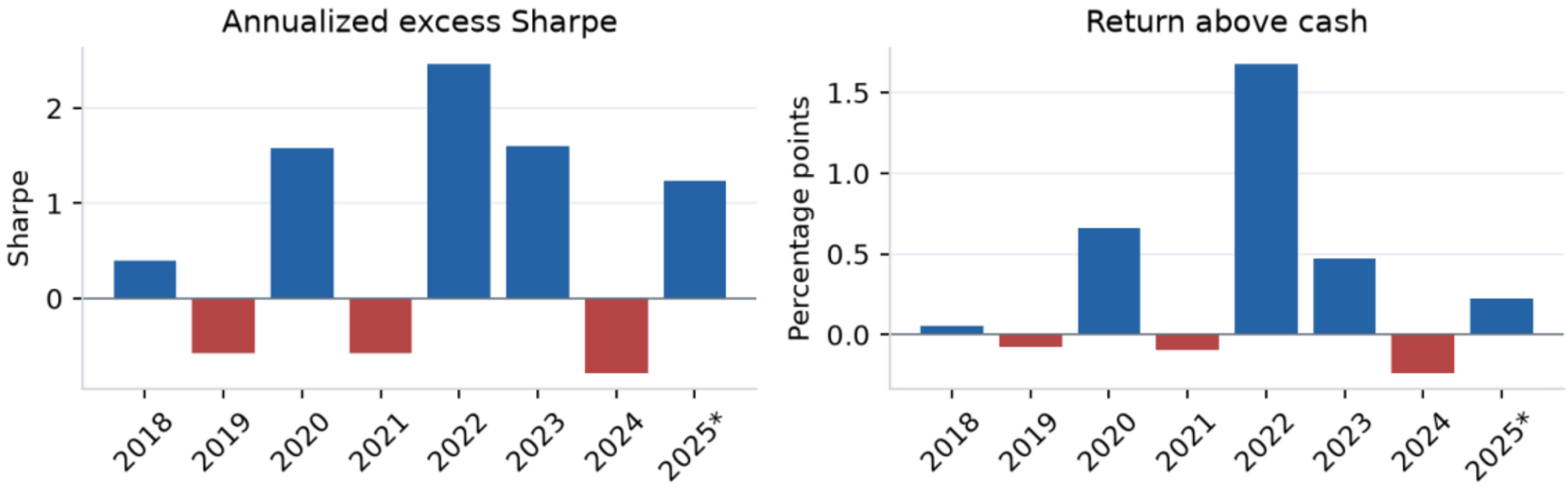
Actual $10m funded portfolios, 2018–29 August 2025. Curves show 100 × (NAV / same-equity cash benchmark − 1), not total NAV returns. All controls share proposals and risk limits; sizing changes admission and inventory.

Sizing	Sharpe	Entries
Fixed 100%	0.769	133
Fixed 50%	0.761	263
Learned 50/75/100%	1.003	212

ANNUAL RESULTS

Profits vary sharply across regimes

Positive cash interest must not conceal trading losses

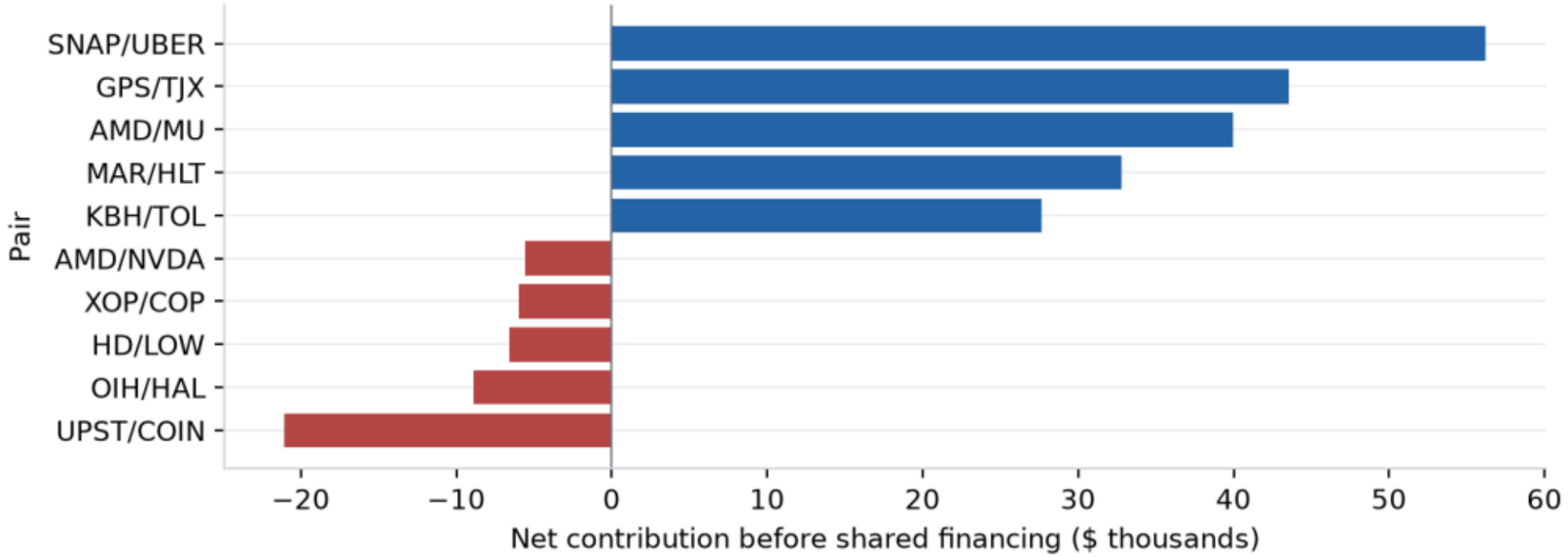


$10m learned strategy. Sharpe is annualized; calendar-period return above cash is not. The 2025 observation is partial through 29 August. Positive total funded returns can coexist with negative trading excess returns.

ATTRIBUTION

Convexity creates both major wins and major losses

Pair contributions in the original research reference, before shared financing

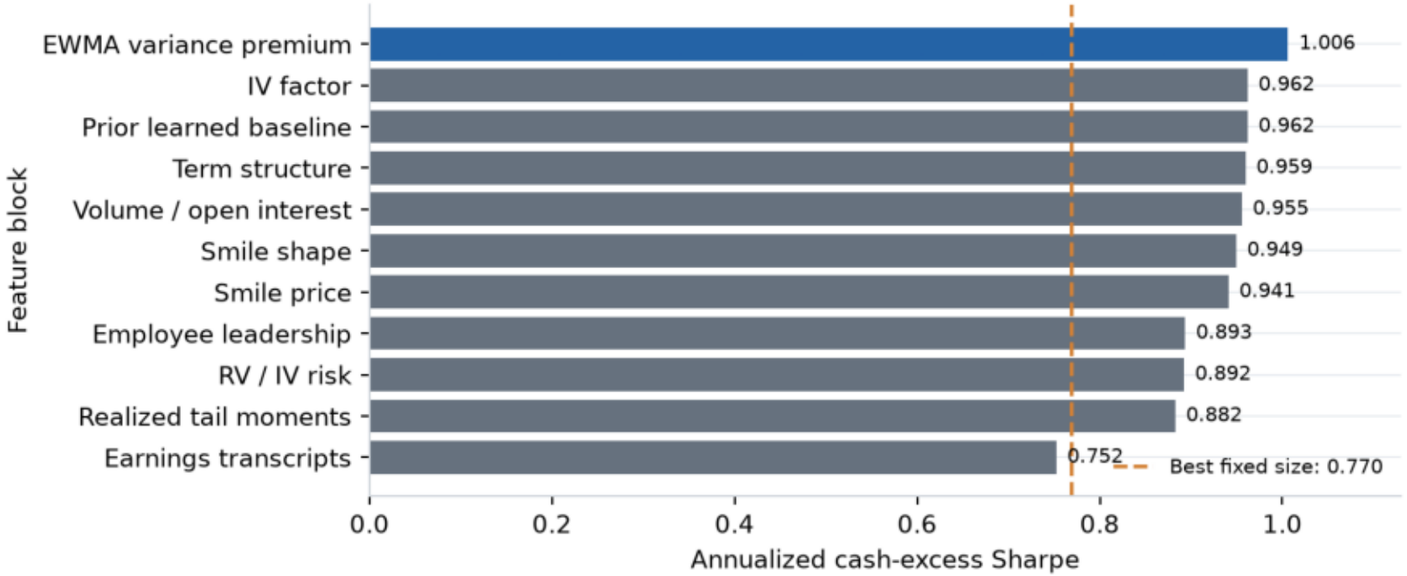


Five largest losing and winning pairs in the original $345m reference. Includes costs, hedges and dividends; excludes shared financing. This is retrospective attribution, not a rule for selecting or banning names.

FEATURE EXPLORATION

More signals did not generally mean higher Sharpe

Frozen proposals; original $345m research reference

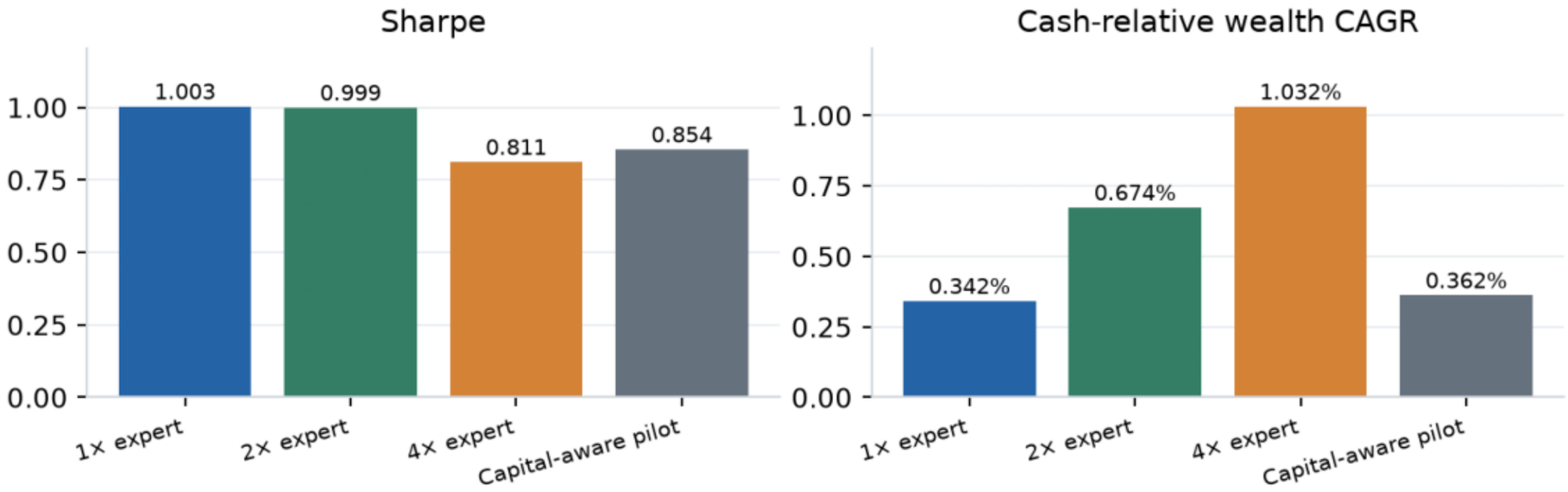


Original $345m research reference, identical 608 qualified proposals. Each block is added separately to the prior 27-input state. The retained variance-premium block reaches 1.006; most additional signals fail to improve the 0.962 learned baseline.

CAPITAL DEPLOYMENT

Simple scaling beats the first capital-aware pilot

All multipliers retain the underlying learned trade sizes



$10m equity with per-pair risk allowances scaled alongside requested trade size. All expert multipliers retain the underlying learned 50/75/100% sizing. Global limits remain fixed. The limited capital-aware pilot underperforms the 1× and 2× expert on Sharpe.

EVIDENCE AND LIMITATIONS

One Sharpe is a research finding, not a live guarantee

What the current experiment establishes—and what it does not

- Learned sizing beats fixed sizing in this sample; current P25 has no fresh supervised-NN comparison.
- 126-session reference bootstrap: Sharpe [0.137, 1.660]; best-fixed gap [0.008, 0.519].
- Intervals are not selection-adjusted; repeated research and retrospective formation remain.
- 2022 concentration, negative 2024, quote-quality and margin/assignment limitations.

CONCLUSION AND NEXT STEP

Preserve the signal; test sequential exits separately

Equity relative-value framework → actual options → learned sizing

- Finding: a downside-put relative-value signal with a persistent historical sizing benefit.
- Next controlled test: identical entries / initial sizes; learn hold versus close both legs.
- Use remaining life, moneyness, relative gamma and event / name inventory.
- Then freeze the design and evaluate untouched dates with prospective pair formation.

APPENDIX

Returns and Sharpe by year at $10m

Funded returns include cash; above-cash figures are percentage points

Year	Funded return	Above cash (pp)	Excess Sharpe
2018	2.139%	+0.056	0.396
2019	2.204%	-0.076	-0.577
2020	1.183%	+0.658	1.581
2021	0.039%	-0.094	-0.580
2022	3.474%	+1.676	2.462
2023	5.328%	+0.472	1.604
2024	5.149%	-0.238	-0.785
2025*	3.263%	+0.225	1.231

*2025 through 29 August; calendar-period returns are not annualized.

REFERENCES

Sources and reproducibility

Updated presentation accompanies a ten-page paper

- Ning & Lee (2024), arXiv:2403.12180
- Ten-page paper (PDF) · Editable paper (HTML)
- Main findings · Funding controls
- Capital-policy pilot · Figure source hashes